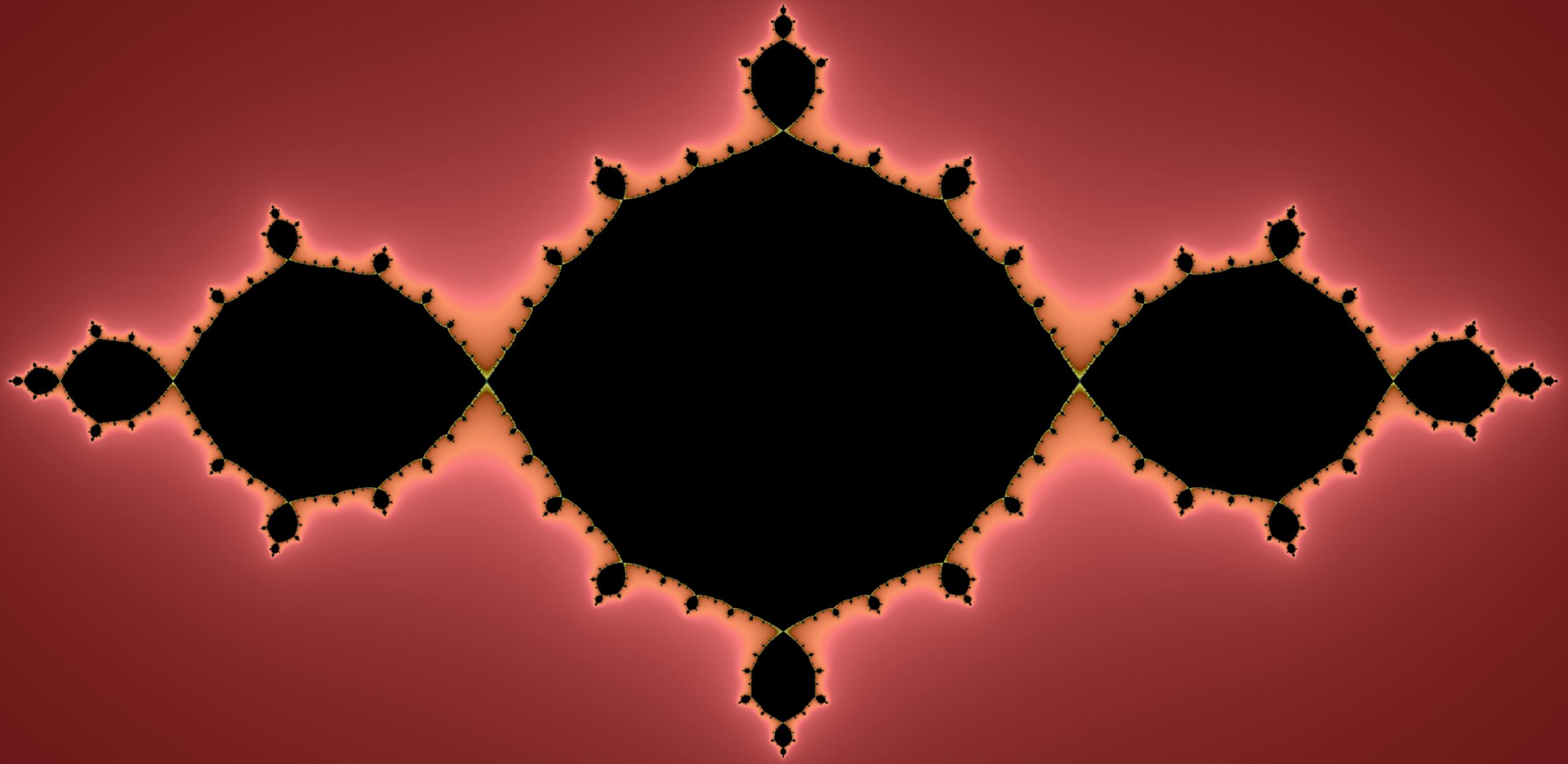
Fractal Software Design – Software Portability and Self Similarity (HAUD)

BECKHOFF



Fractal Software Design

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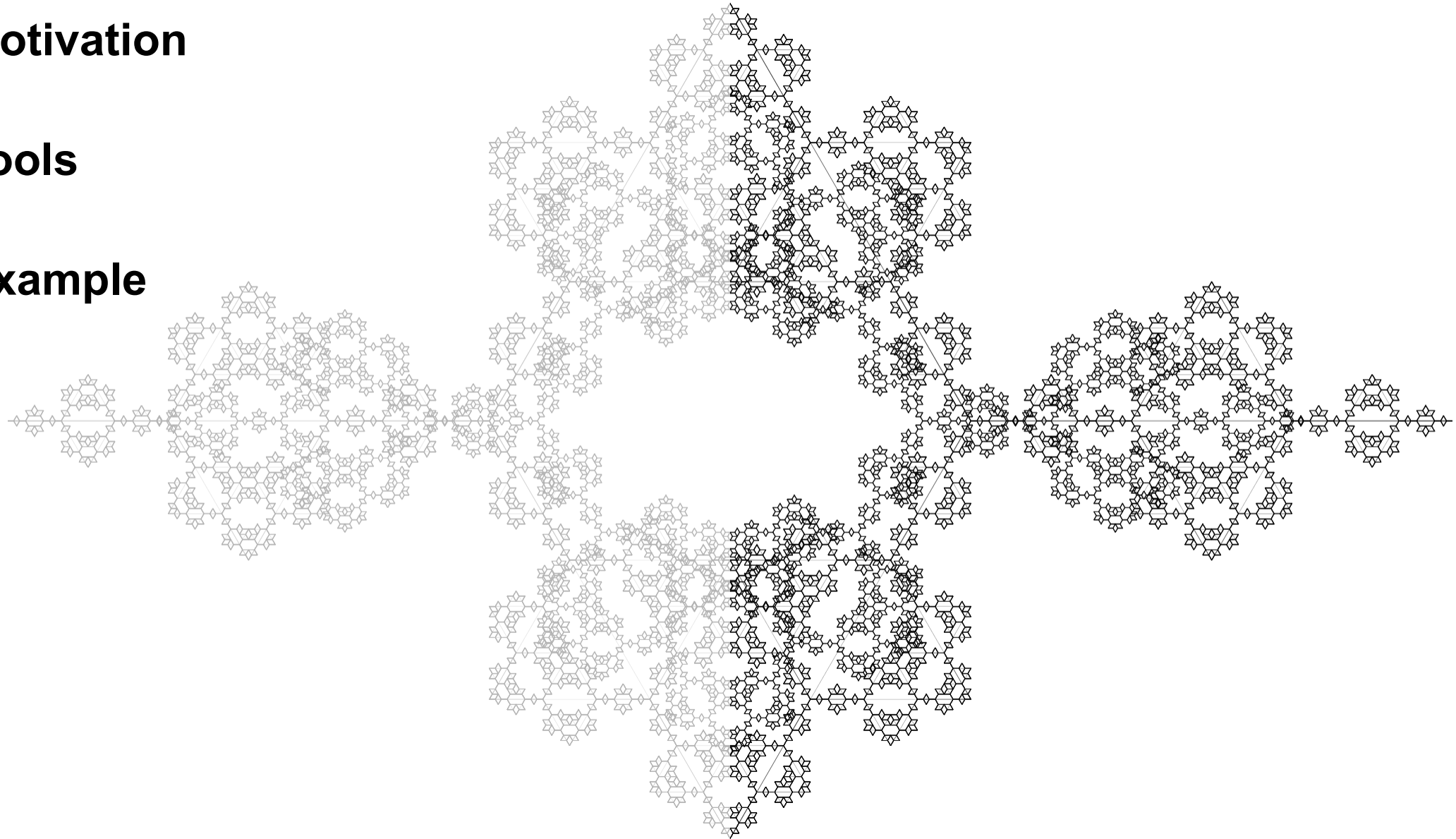


1. Motivation

- **Software portability**
  - Top level business logic shall not suffer from changes in lower abstraction levels.
- **Rigid hardware abstraction**
  - Unified software mechanics to shield business logic from hardware variance.
- **No God Objects**
  - Edge cases shall not disrupt abstraction.
- **Same cycle responsiveness**
  - Eliminate traditional, monolithic state machine design flaws.

1. Motivation

- **Software portability**
  - **Separation of business logic and motion logic enables portability.**
  - **Changes in motion layer:**
    - Adding new features does not break the code.
    - Swapping a hardware controller does not break your business logic.
    - Introducing a new motion kernel does not break the code.

1. Motivation

- **Hardware Abstraction Layer**
  - **Strict interfaces enforce behavioural contract.**

    Rigid = Enforced Contract = Safety

  - **Rigid Interfaces ensure:**
    - Business logic never couples to hardware.
    - Hardware changes do not affect business logic.
    - Different hardware can be swapped transparently.

1. Motivation

- **NO God Objects**
  - **Work local, ship global.**
    - **Composition over inheritance.**
  - **Composition ensures:**
    - Specialized logic can be reused.
    - Scales easily with future development.
    - Know Thyself:
      - Know your job and do it well

1. Motivation

- **Same Cycle Responsiveness and Cycle Optimal Programming**
  - **Break the cycle.**
    - **Find a reusable structure to break safely.**
  - **Logic progression:**
    - Do not waste time to switch decision trees.
    - Ensure observability.
    - Keep decisions simple

1. Motivation

- **WHY?**
  - **Moving at 10 m/s requires cycle optimal programming.**
  - **Synchronizing place and time with software that is fast and safe.**
  - **The machine must observe itself and tell the user about it.**
  - **Communication agnosticism is a requirement for distributed motion applications.**
  - **Top application layer must be oblivious to the hardware executing the commands.**

2. Tools

- **SOLID DESIGN**
  - **S — Single Responsibility Principle**
    - a class should have one, and only one, reason to change.
  - **O — Open/Closed Principle**
    - software entities should be open for extension but closed for modification.
  - **L — Liskov Substitution Principle**
    - Objects of a superclass shall be replaceable with objects of its subclasses without breaking the application.
  - **I — Interface Segregation Principle**
    - users should not be forced to depend upon interfaces that they do not use.
      - a software entity only sees the methods it actually needs to do the work.
  - **D — Dependency Inversion Principle**
    - Depend upon abstractions, not concretions.

2. Tools

- **ACID-inspired principles for cyclic RTS**
  Borrowed from database transaction properties and adapted to deterministic control logic.
  - **A — Atomicity**
    - **Within a cyclic real-time task, a bounded local logic step can be treated as one deterministic, observable operation.**
  - **C — Consistency**
    - **A transaction must bring the system from one valid state to another valid state.**
      - **The required handshake sequences perform a consistent execution. The behaviour is fully testable.**
  - **I — Isolation**
    - **Concurrent transactions should not interfere with each other.**
      - **The architecture provides structural isolation by design.**
  - **D — Durability**
    - **Once a transaction has been committed, it will remain so, even in the event of power loss, crashes, or errors.**
      - **The loss of state or information must lead to a coordinated recovery.**

2. Tools

- **Clarifications**
  - The **SOLID** tools are directly applicable in TwinCAT OOP.
    - The decision how deep inheritance runs is influenced by the task at hand.
    - In my world, a flat inheritance tree has certain advantages regarding variations and the taming of complexity.
      - Since there is no “one size fits all”
        - → the depth of inheritance is up for consideration
    - Base classes convey the intent of custom extensions that solve an adjacent problem or substitute an existing implementation without changing the code.
    - Interfaces are used as horizontal connector for classes that might share common properties.

2. Tools

- **Clarifications**
  - The **ACID** tools require some explanations.
    - Transactional and relational principles share properties that are also applicable to PLC programming.

      - **Atomicity (Logic)**: The cyclic real-time task provides a deterministic execution boundary for bounded local logic. Within that boundary, internal decisions can be executed as one observable logical step.

        - We now only have to compress the traditional state machine changes (that consume a PLC cycle) into one observable sequence that executes without cycle change. By employing such sequential state execution, the logic needs only to wait for feedback when it is required.

        - Further is it helpful to eradicate the looping of arrays and classic buffers
          - Preallocated, bounded linked structure can avoid array shifting. „GetHead“ and „AddTail“ provide O(1) FIFO operations.

          - Preallocation prevents runtime memory allocation and fragmentation risk.

2. Tools

- **Clarifications**
  - The **ACID** tools require some explanations.
    - A lot of similar problems are coded in vastly different ways, mostly caused by crunching time during FAE.

      - **Consistency (architecture)**: The facade pattern as helper
        - If everything works the same, reading the code is made easier.

        - Transactional handshakes have two trade offs
          - Easy to read
          - Easy to maintain

        - A **consistent** use of the facades for handshakes:
          - preserves defined machine invariants
          - provides one observable outcome for every command:
            - **accepted, rejected, executing, completed, aborted or faulted.**

2. Tools

- **Clarifications**
  - The **ACID** tools require some explanations.
    - Preventing unintended or unauthorized modification of validated cyclic logic and state is essential.

      - **Isolation (architecture)**: Again; the facade pattern as helper, and the horizontal interface pointer as guard.
        - If everything works an addition or deletion of functionality somewhere else must not break the architecture.

        - Isolation by design is the best shield
          - Semaphores in a cyclic RTS system add shared synchronization state, timing risks, and maintenance obligations. Prefer explicit ownership and cyclic handover wherever possible.

        - Each mutable Ctrl / State facade has one defined cyclic owner. Other tasks, mappings, ADS clients, and application layers submit requests through the facade and do not directly alter internal cyclic state.

        - Architectural isolation prevents unwanted dependencies
          - ownership rules prevent conflicting writers.

2. Tools

- **Clarifications**
  - The **ACID** tools require some explanations.
    - Saving data is one thing, saving machine state another.

      - **Durability (design & physical constraint)**: On the topic of a machine, it is vital that machine parts are not colliding in the physical world.
        - Fetch the data you need for recovery in time.
          - Use UPS for your IPC → save persistent data
          - Use the request for interruption ahead of any physical reaction from the machine to save the data

        - Data after recovery is the last known logical state, **not** guaranteed physical reality.
          - Before motion is enabled after restart, recovery must evaluate data regarding safety status, drive/axis/mover state, position and reference validity, and process feedback.

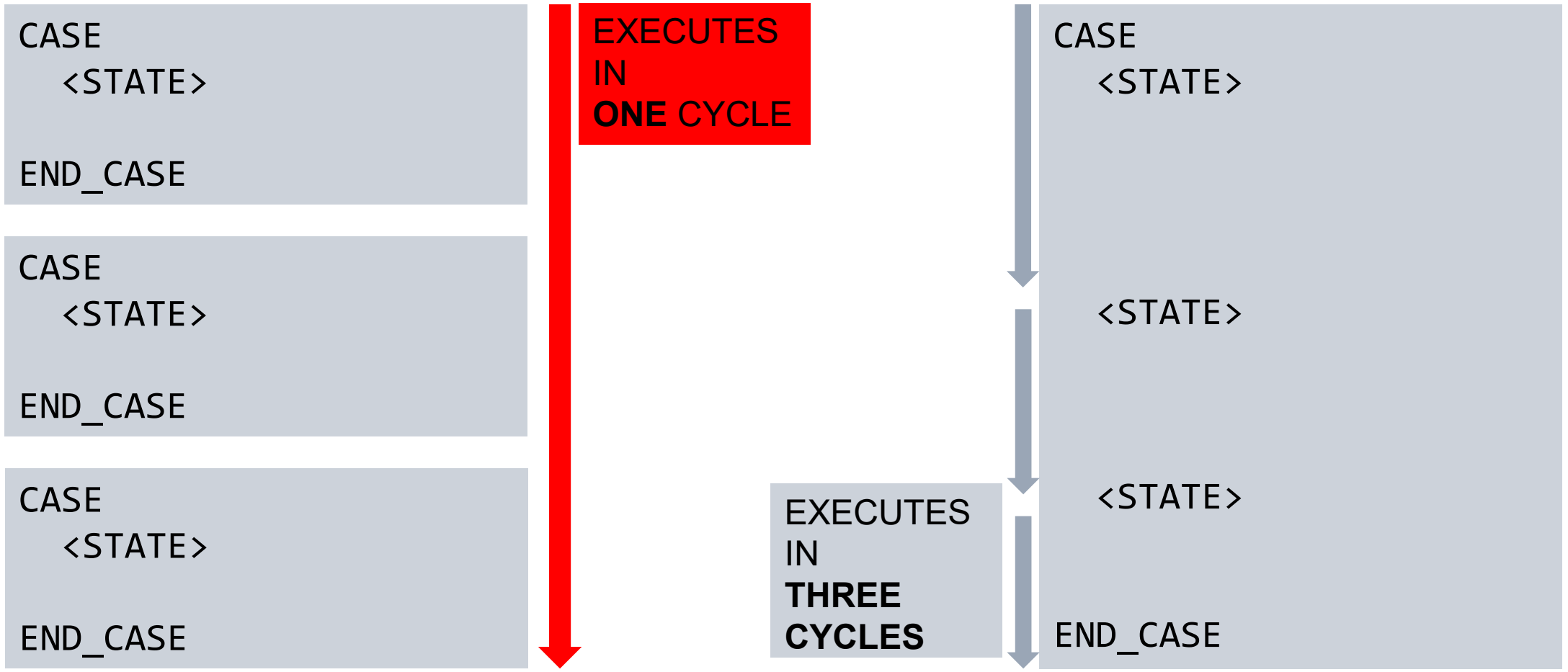
        - Design a recovery strategy for the machine
          - A generic solution is either not working or complex as hell.

3. Examples

- **Break the cycle between states with E_PROGRESS.**
  - Core feature of the required Same Cycle Responsiveness.
  - Ensure logical atomicity for bounded local state progression in the cyclic RTS.
    - Logical atomicity applies to local decisions and coherent command publication. Physical effects are asynchronous and cannot generally be rolled back.
  - Ensure machine readability without having to construct machine semantics on top.
  - Ensure human readability without translation of numbers into text.
  - Time Optimal Programming while keeping source code readable.

3. Examples

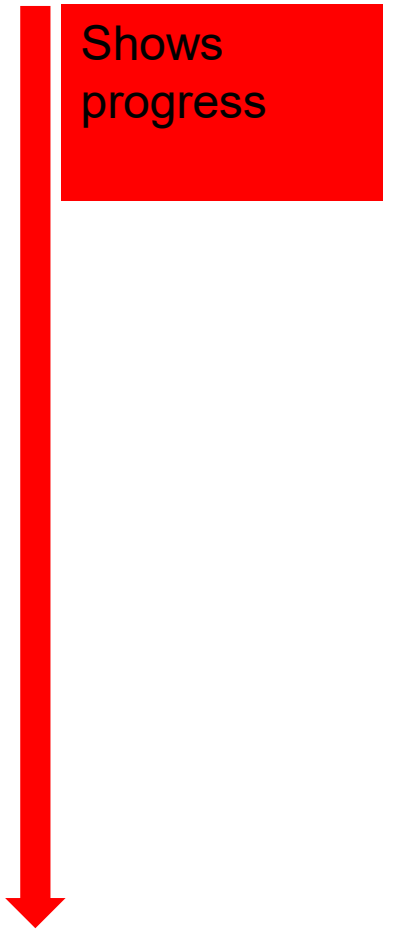
– **Break the cycle between states with E_PROGRESS.**



3. Examples

– **Break the cycle between states with E_PROGRESS.**

```
{attribute 'to_string'}
TYPE E_PROGRESS :
(
  // progress has 2 use cases
  // 1. as offset to cyclic interface's state
  //    for the requested command/function
  //    e.g. State := <Enum equivalent to eCmd> + E_PROGRESS
  //
  // 2. as state feedback (result) from (any) method
  PROGRESS_INVALID,
  PROGRESS_NOT_EXIST      :=   100,
  PROGRESS_INIT           :=  1000,
  PROGRESS_BUSY           :=  2000,
  PROGRESS_PREPARE        :=  3000,
  PROGRESS_STARTUP        :=  4000,
  PROGRESS_CHECK          :=  5000,
  PROGRESS_OCCUPIED       :=  6000,
  PROGRESS_WORKING        :=  7000,
  PROGRESS_STILL_WORKING  :=  8000,
  PROGRESS_ERROR          :=  9000,
  PROGRESS_DONE           := 10000
)UINT;
END_TYPE
```

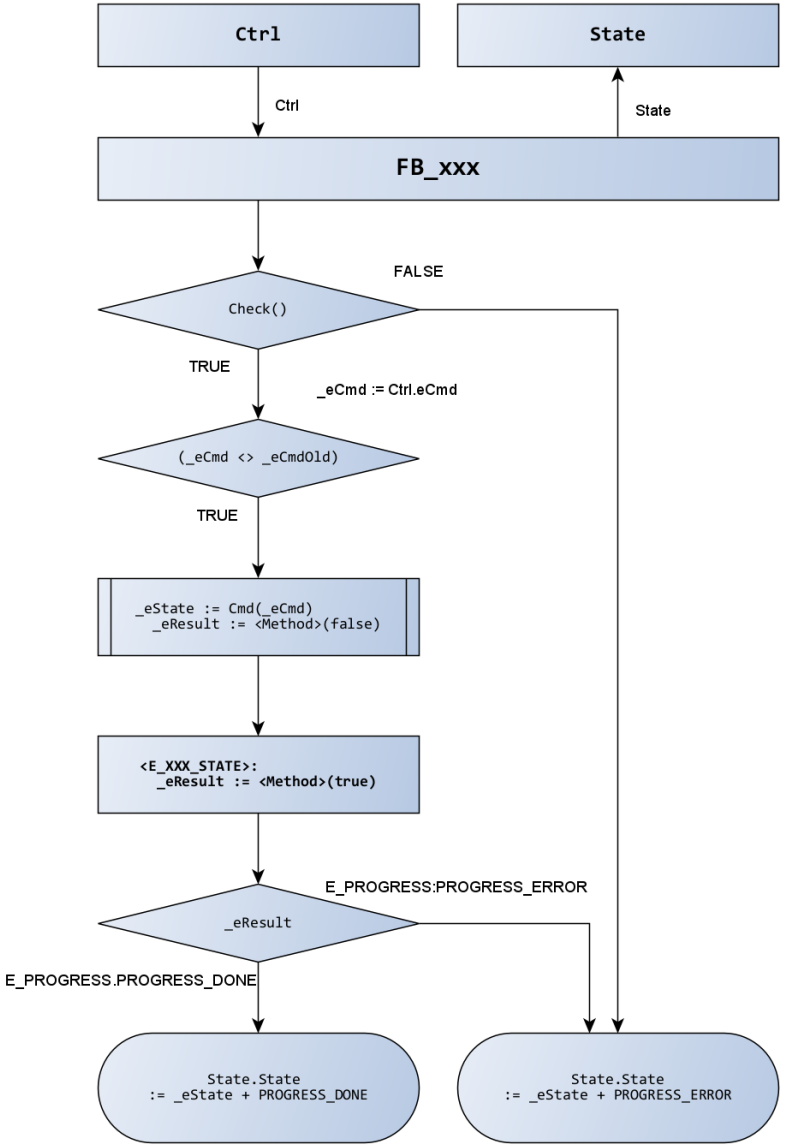


3. Examples

- **The Facade:**

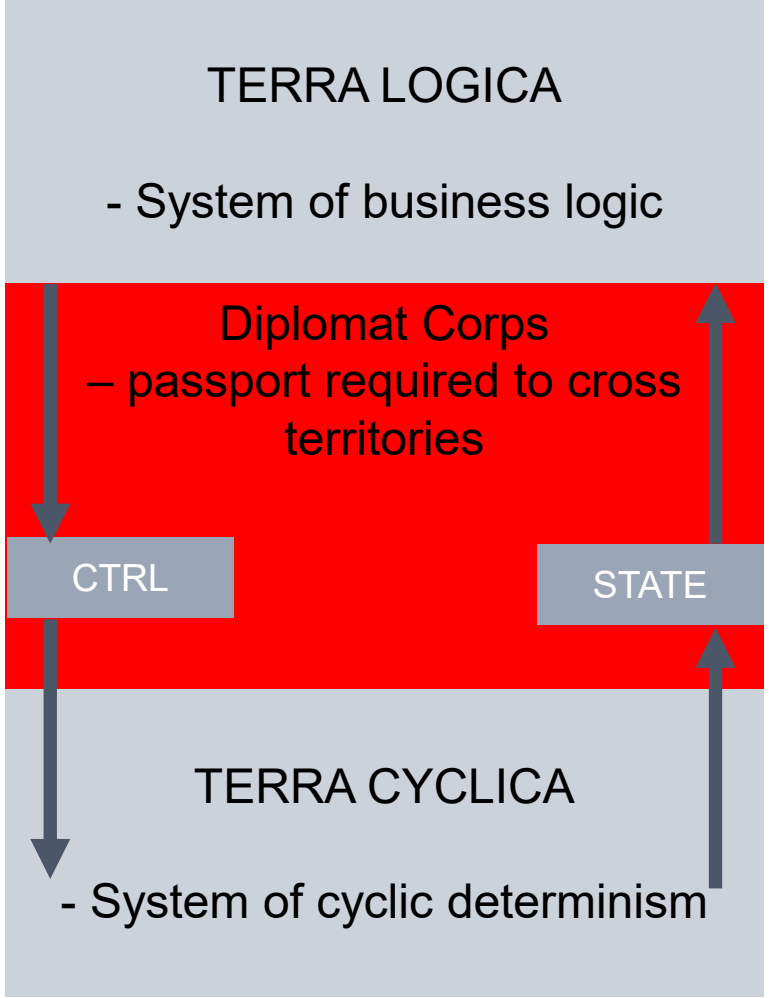
- State / Ctrl structures
  - Establishes unified access
  - Commands can simply be ‘dropped’ into Ctrl datafield by its owner
    - Enables asynchronous communication
    - Enables cyclic communication mapped onto any cyclic fieldbus
    - State is updated by own cyclic call

- State feedback for cyclic class wrappers
  - **Always** combined with E_PROGRESS
    - You can filter your response by a simple modulo division



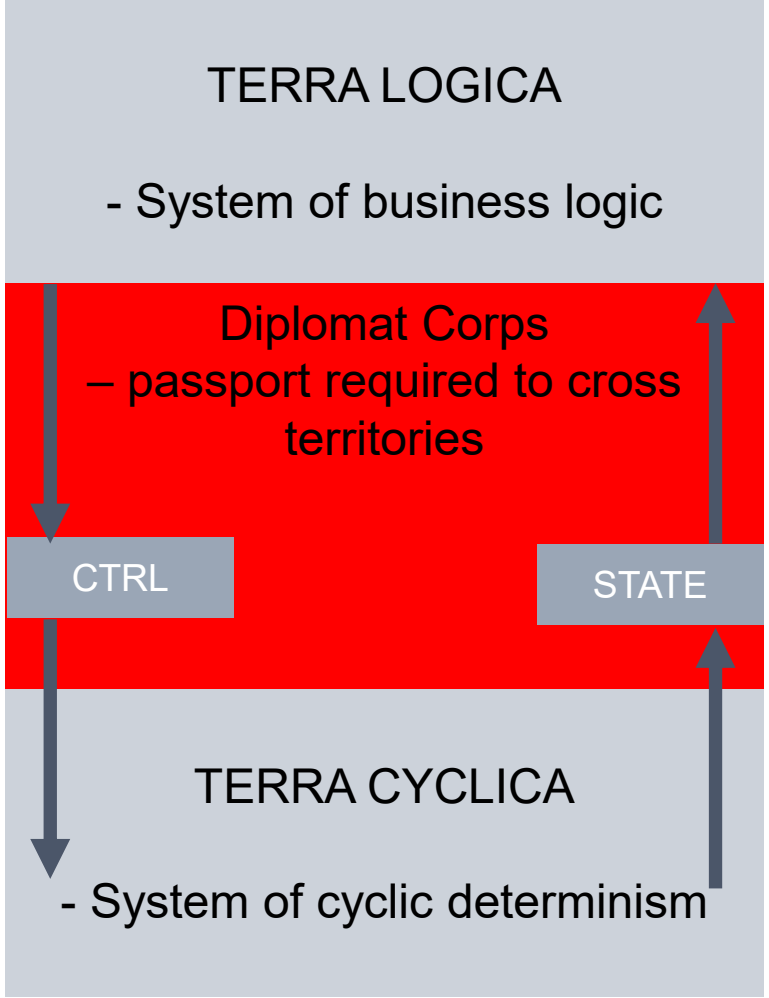
- **Diplomats and Territories - a tale of two systems**

- **This world view depends on discipline not compiler restraints.**
- **The architecture operates on the principle of strict namespace encapsulation.**
- **The core TcGVL (Global Variable List) files are treated as sovereign territory.**
- **No external logic is permitted to read or write to this territory directly.**

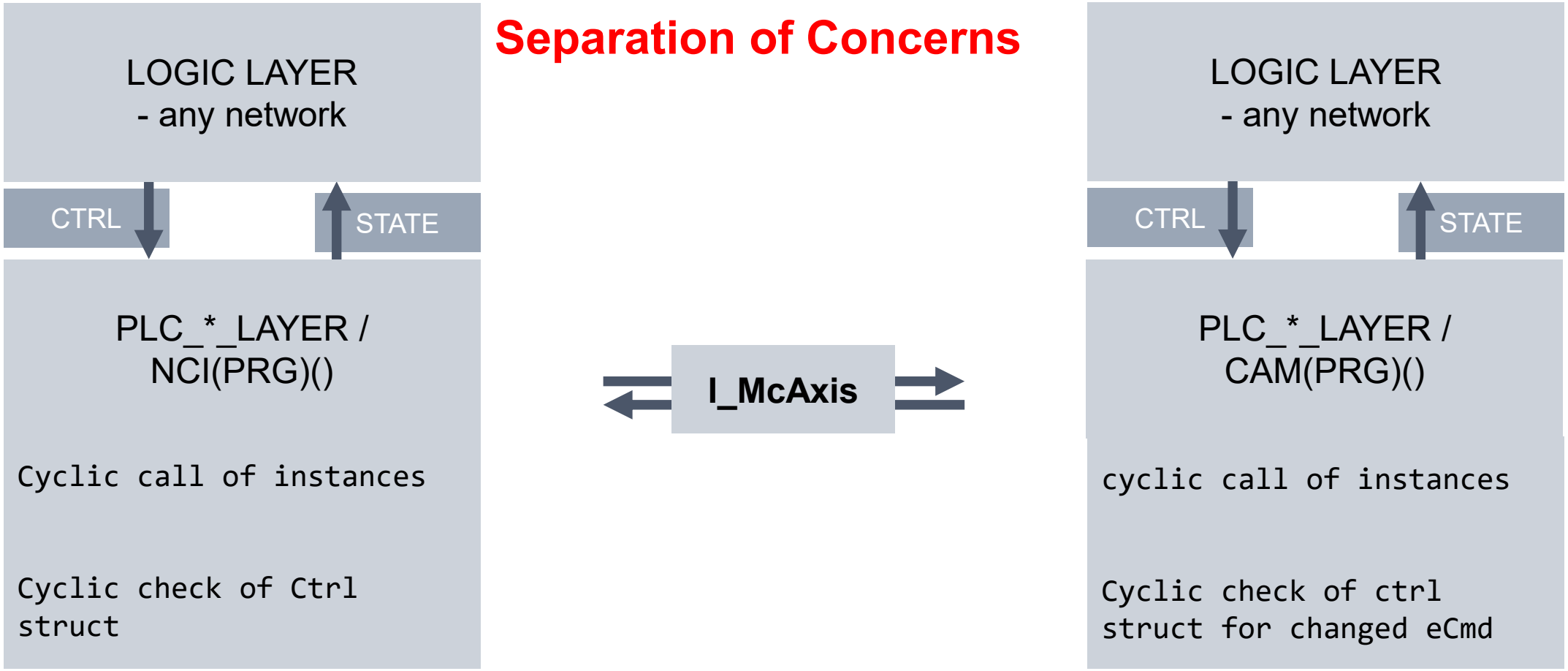


- **Diplomats and Territories - a tale of two systems**

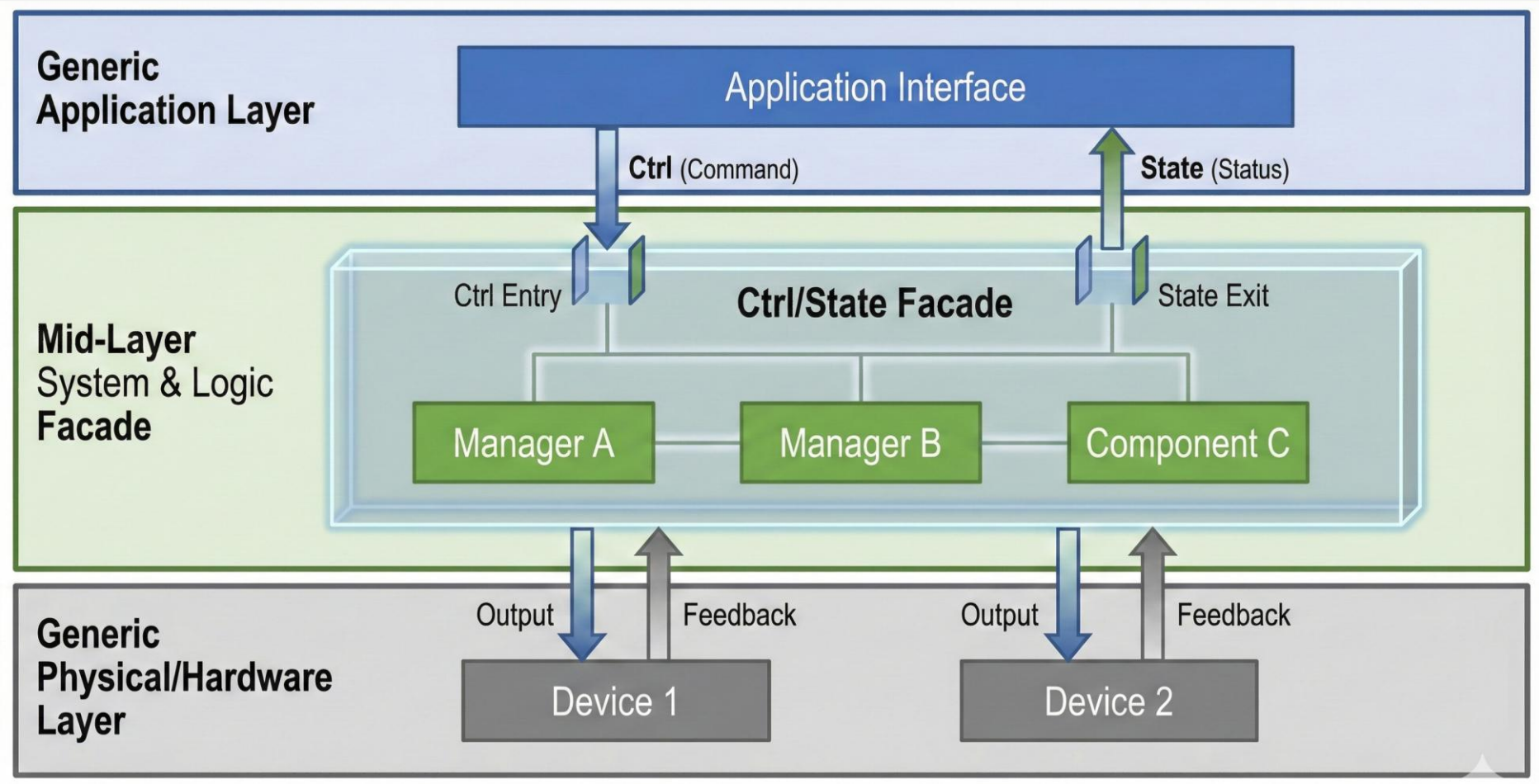
- **Instead, communication requires a "diplomat"—a strictly defined Facade or Interface.**
- **Any cyclic application may be connected via MAPPING procedures (fieldbus, EAP).**
- **Any non-cyclic application may be connected via symbolic access.**
- **Hybrid access possible**
  - **A middleware layer to organize access (switching between ADS and cyclic mapping) keeps the base layer unchanged**



- **Software Design Example:**

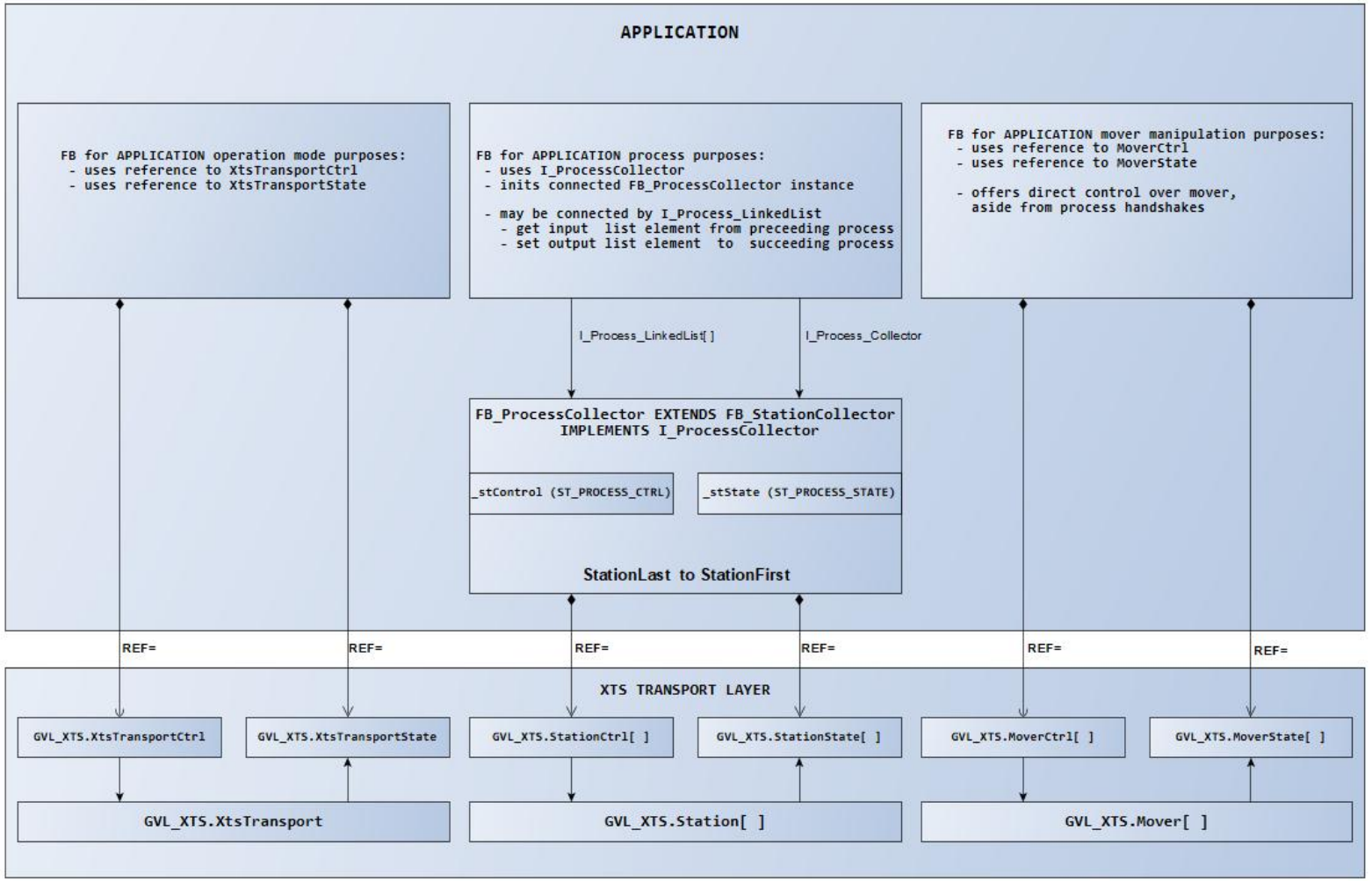


- **Software Design Example:** **The Facades**



- **Software Design Example:**
  - **Build applications layer by layer**
  - **The transport logic lives in instances**

The Facades



- **Software Design Example:** **The Facades**
  - **MAIN(XtsTransport)**
    - MAIN — application orchestration and cyclic instance calls

```
// cyclic call
GVL_XTS.XtsTransport();

// main control
GVL_XTS.XtsTransport.Ctrl                   REF= GVL_XTS.XtsTransportCtrl;   // main control
GVL_XTS.XtsTransport.State                  REF= GVL_XTS.XtsTransportState;  // main state

// member controls
GVL_XTS.XtsTransport.XpuCtrl                REF= GVL_XTS.XpuCtrl;
GVL_XTS.XtsTransport.XpuState               REF= GVL_XTS.XpuState;
GVL_XTS.XtsTransport.XpuInfo                REF= GVL_XTS.XpuInfo;

GVL_XTS.XtsTransport.GroupItf               :=   GVL_XTS.CaGroupItf;
GVL_XTS.XtsTransport.GroupInfo              REF= GVL_XTS.CaGroupInfo;

GVL_XTS.XtsTransport.MoverItf               REF= GVL_XTS.MoverItf;
GVL_XTS.XtsTransport.MoverInfo              REF= GVL_XTS.MoverInfo;
GVL_XTS.XtsTransport.MoverLastPosition      REF= GVL_XTS.LastPosition;
GVL_XTS.XtsTransport.MoverLastGap           REF= GVL_XTS.LastGap;

GVL_XTS.XtsTransport.StationStartIndex      :=   GVL_XTS.StationStartIndex;
GVL_XTS.XtsTransport.StationStart           REF= GVL_XTS.StationStart;      // start position for E_XTS_TRANSPORT_CTRL.CMD_TRANSPORT_START
GVL_XTS.XtsTransport.StationCtrlItf         REF= GVL_XTS.StationCtrlItf;    // station interfaces are required for accessing station methods
GVL_XTS.XtsTransport.StationListsItf        REF= GVL_XTS.StationListItf;    // list interface is required for access to linked list methods
GVL_XTS.XtsTransport.StationControl         REF= GVL_XTS.StationCtrl;
GVL_XTS.XtsTransport.StationState           REF= GVL_XTS.StationState;
```

- **Software Design Example:**
  - **GVL_APPLICATION (ExternControl)**
    - GVL_APPLICATION — application-facing command facades and instance declarations.

The Facades

GVL_APPLICATION | MAIN_APP

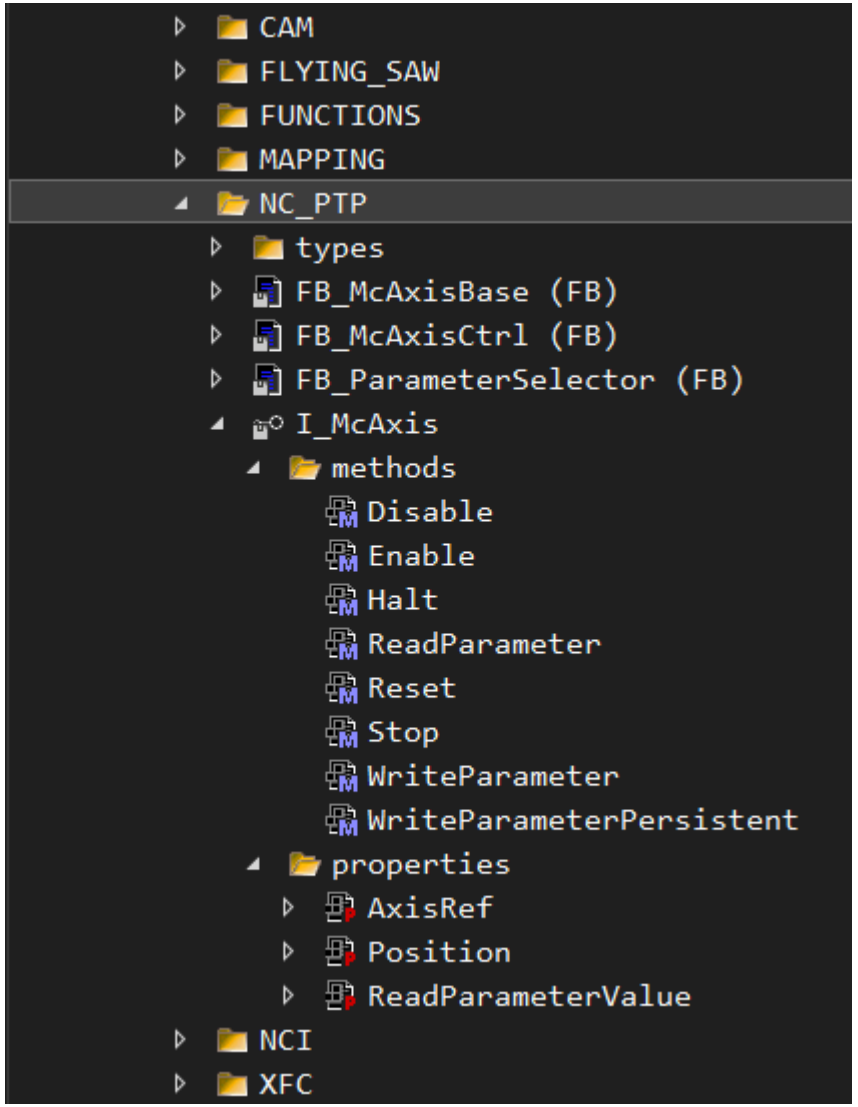
```
VAR_GLOBAL CONSTANT
  MAX_PROCESS         : E_INSTANCE := E_INSTANCE.SENDER_BUFFER_INFEED;
END_VAR
{attribute 'qualified_only'}
VAR_GLOBAL
  //-------------------------------------------------------------------------------------------------
  // Application
  //
  //  this is the part you will have to change and adapt into your programming
  //-------------------------------------------------------------------------------------------------
  //-------------------------------------------------------------------------------------------------
  // fb_Application:
  // - procedures for getting fb_Instance classes and XTS_TRANSPORT_LAYER to work
  //
  //  - Cmd/state behaviour similar to sw mechanics in fb_TransportUnit
  //  - E_INSTANCE_CMD  / E_INSTANCE_STATE + E_PROGRESS
  //      INIT          /   INIT          : initialize processes and XtsStations
  //      DISABLE       /   DISABLED      : RemoveAllAxis from CA Group, disable CA Group, disable all movers
  //      ENABLE        /   ENABLED       : AddAllAxis to CA Group, Enable CA Group, Enable all movers
  //      WORK          /   WORKING       : CMD_TRANSPORT_START, enable handshakes, start writing list entries to BUFFER_INFEED
  //      PROCEED       /   PROCEEDING    : CMD_TRANSPORT_RESTART, enable handshakes, continue writing list entries to BUFFER_INFEED
  //      FINISH        /   FINISH        : stop writing list entries to BUFFER_INFEED, handshake processes until outfeeds are empty
  //      CLEAR         /   CLEARED       : clear all instances of fb_Instance, clear all XtsStations
  //-------------------------------------------------------------------------------------------------
  ApplicationMain       : fb_Application;
  AppControl            : ST_APP_CTRL;
  AppState              : ST_APP_STATE;

  //-------------------------------------------------------------------------------------------------
  // fb_Instance control and state
  //  - structs are used in fb_Application as reference
  //-------------------------------------------------------------------------------------------------
  InstanceCtrl          : ARRAY[1..MAX_PROCESS] OF ST_INSTANCE_CTRL;
  InstanceState         : ARRAY[1..MAX_PROCESS] OF ST_INSTANCE_STATE;

  TicketListControl     : ST_LIST_CTRL;
  TicketListState       : ST_LIST_STATE;
  TicketListDataInput   : ST_LIST_NODE_DATA;   // datafield for you to write(add) your data
  TicketListDataOutput  : ST_LIST_NODE_DATA;   // datafield for you to read (get) your data
END_VAR
```

- **Software Design Example:** **The Facades**
  - **GVL_XTS(XtsTransport)**
    - GVL_XTS – transport-facing structures and instance declarations.

```
26
27      // ctrl / state datafields for extern control
28      StationCtrl         : ARRAY[1..MAX_STATION] OF ST_STATION_CTRL;   // control for extern to write, you can do whatever you want as long as you do as I say
29      StationState        : ARRAY[1..MAX_STATION] OF ST_STATION_STATE;  // state for extern to read, this is what I tell you to do
30
31      // station parameter
32      StationParameter    : ARRAY[1..MAX_STATION] OF ST_STATION_PARAMETER; // wherever, whenever you want. Build it and they will come...
33
48
49      XtsTransportCtrl    : ST_XTS_TRANSPORT_CTRL;    // control of XTS transport for extern to write
50      XtsTransportState   : ST_XTS_TRANSPORT_STATE;   // state   of XTS transport for extern to read
51
81
82      //-------------------------------------------------------------------------
83      // MC2 encapsulation, CA movement
84      Mover               : ARRAY[1..MAX_MOVER] OF fb_MoverCtrl;    // MC and CA methods for mover; cyclic check for individual command on mover
85
86      MoverCtrl           : ARRAY[1..MAX_MOVER] OF ST_MOVER_CTRL;   // control of individual mover for extern to write
87      MoverState          : ARRAY[1..MAX_MOVER] OF ST_MOVER_STATE;  // state   of individual mover for extern to read
88
89      MoverSelector       : fb_MoverSelector;                       // alternative for moving a single mover via Index/ID selection
90      MoverSelectorCtrl   : ST_MOVER_SELECTOR_CTRL;
91      MoverSelectorState  : ST_MOVER_SELECTOR_STATE;
```

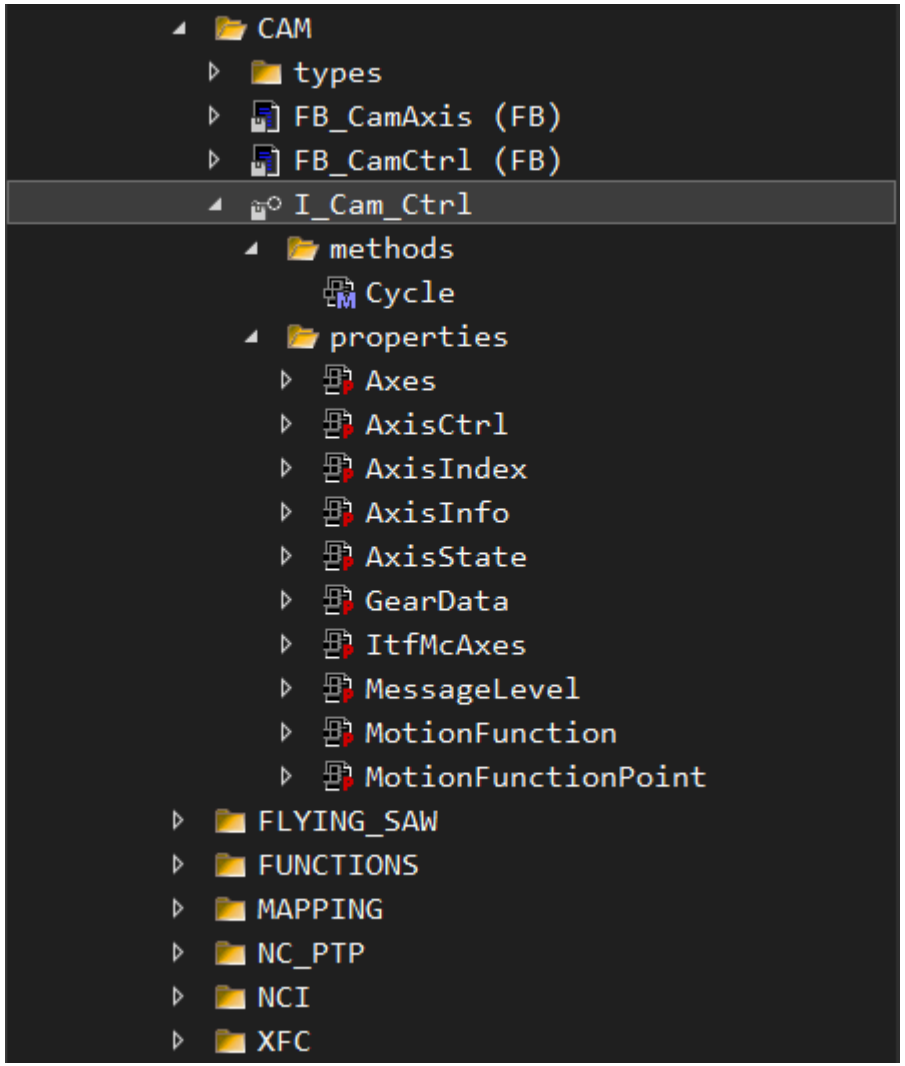
- **Software Design Example:**
  - **Distributes only required methods and properties.**
  - **Enables strict separation of concerns and connects classes horizontally**
  - **I_McAxis is the base interface and is not optional.**

The Interface



- **Software Design Example:**
  - **Distributes only cycle method. The commands are written onto the facade structures.**
  - **Enables use of empty interfaces for pre-compile defines**
  - **Shall be void of any library datatypes from functions that are optional.**

The Interface



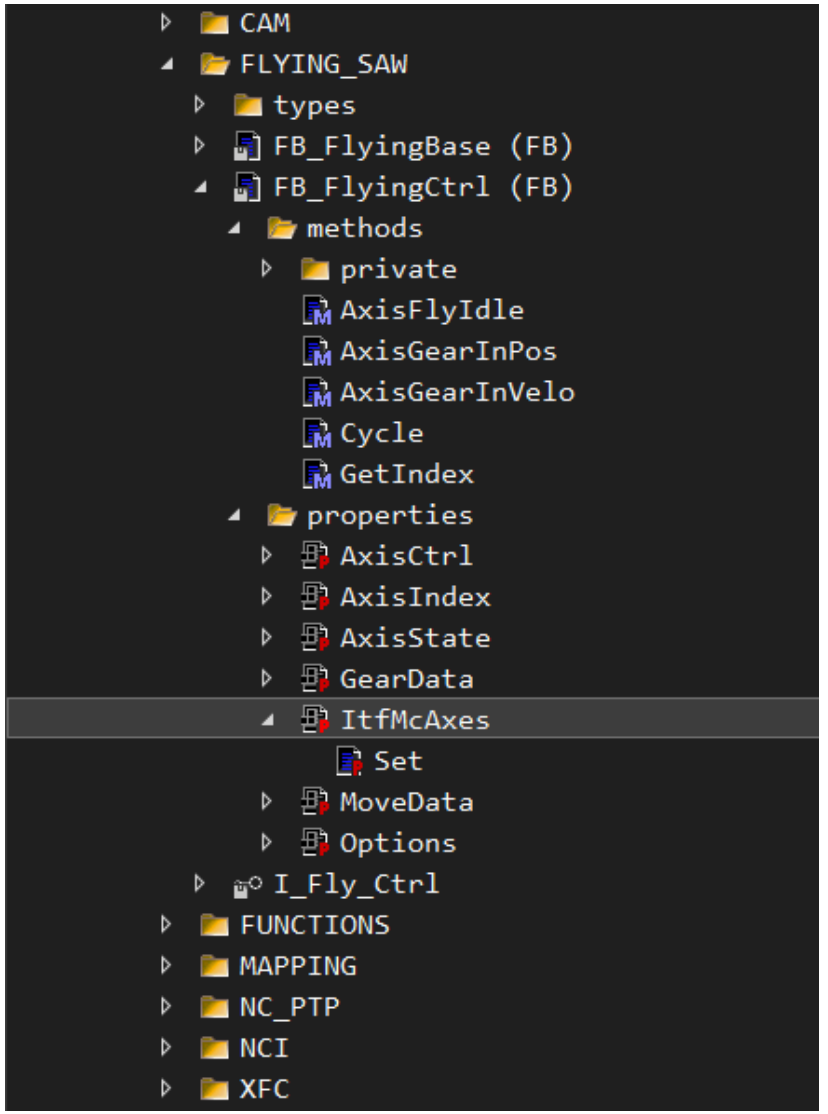
- **Software Design Example:**
  - **The empty interface**
  - **pre-compile defines**
  - **Optional binding is possible with this**
  - **The codebase does not need to change**

The Interface

```
  //-----------------------------------------------------------------------
  // cyclic interface function block: FB_CamCtrl
  //   - MC_Camming functions are here
  //-----------------------------------------------------------------------
{IF defined (CAM)}
  {info 'PLC_MOTION_LAYER: CAM'}
  Control                    : ARRAY[1..PLC_CONSTANT.MAX_AXIS] OF FB_CamCtrl;
{ELSE}
  Control                    : ARRAY[1..PLC_CONSTANT.MAX_AXIS] OF I_Cam_Ctrl;
{END_IF}
END_VAR
```

- **Software Design Example:**
  - **Injected in class via property**
  - **Class is oblivious to implementation details**
  - **This is the contract on offer**

The Interface



- **Public on GitHub:**
  - https://github.com/haud-ba/PLC_MOTION_LAYER
  - https://github.com/haud-ba/XTS_TRANSPORT_LAYER
  - https://github.com/HAUDX/XTS_TRANSPORT_EXAMPLES